A Comparative Study of Text Representations for News Classification

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Abstract

This report compares several text representation methods for news classification on the New York Times (NYT) dataset. We evaluate sparse lexical representations including binary bag-of-words, word frequency, and TF–IDF, as well as dense document representations obtained by averaging 100-dimensional GloVe and Word2Vec embeddings. Word2Vec models are trained on either AG News or NYT corpus. Finally, we fine-tune the pretrained `BERT-base-uncased` model for three epochs with a maximum sequence length of 64 tokens. Logistic Regression is adopted for all feature-based baselines. All experiments reuse a fixed 80/10/10 train/validation/test split, and model performance is measured on held-out NYT test set using Accuracy and Macro-F1.

1 Introduction

Text classification assigns a predefined category to a document. News classification is a useful setting for studying the effect of text representations because news articles contain both topic-specific vocabulary and contextual information. This report compares three families of representations: sparse bag-of-words features, averaged static word embeddings, and a contextual pretrained language model.

The experiments are designed to isolate the contribution of representation choice. Logistic Regression is held fixed for the sparse and embedding-based approaches, while BERT is fine-tuned end-to-end for classification. The main research questions are: (1) how do binary and frequency-based lexical features compare; (2) how do pretrained and corpus-trained static embeddings compare; (3) how does contextual fine-tuning compare with the feature-based baselines?

2 Dataset and Experimental Setup

2.1 Datasets

The NYT dataset is the primary classification dataset and contains two fields: *text* for news content and *label* for news category. It contains 11447 news documents in total with three target categories: *sports*, *politics* and *business*. After random shuffling, we obtain 9157 training samples, 1145 validation samples and 1145 held-out test samples. The dataset is heavily imbalanced: the *sports* class dominates with 8597 documents, whereas *politics* and *business* contain only around 1400 samples each. AG News corpus is only used for training Word2Vec embeddings and never participates in classification training or evaluation.

2.2 Data Split

NYT examples are shuffled and split into training, validation, and test partitions using an 80%/10%/10% ratio. The same split is reused across all methods to guarantee fair comparison. Feature vocabularies, Logistic Regression models, and NYT-trained Word2Vec are fitted **using training data only** to avoid test-set leakage. The test set is held out and only used for final evaluation.

2.3 Evaluation Metrics

We report Accuracy and Macro-F1. Accuracy is the proportion of correctly classified examples:

$$\text{Accuracy} = \frac{\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i]}{N}. \tag{1}$$

Macro-F1 computes F1 separately for each class and averages the class-level scores equally:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c. \tag{2}$$

Macro-F1 is critical under class-imbalanced scenarios since every class contributes equally to the

final score, preventing the overall accuracy from being overly optimistic due to majority classes.

3 Methods

3.1 Bag-of-Words Features

We train Logistic Regression classifiers using the following lexical representations.

Binary BoW. Each document is represented by a sparse vector whose value is 1 if a vocabulary term occurs in the document and 0 otherwise. This representation captures term presence but ignores repetition.

Word Frequency. Each vector dimension records the raw count of a vocabulary term in the document. Unlike binary BoW, this representation preserves repeated word occurrences.

TF–IDF. As an additional frequency-normalized baseline, TF–IDF weights terms by their within-document frequency and discounts terms that appear across many documents. The vocabulary and inverse-document-frequency statistics are fitted on the training split only.

3.2 Averaged Static Word Embeddings

Each document is represented by the arithmetic mean of the embeddings of its in-vocabulary tokens. For a document with token embeddings $e(w_1), \ldots, e(w_n)$, its representation is

$$\mathbf{v}(d) = \frac{1}{n} \sum_{i=1}^{n} e(w_i). \tag{3}$$

Tokens absent from the embedding vocabulary are ignored. If a document has no in-vocabulary tokens, a zero-vector fallback is applied consistently.

Pretrained GloVe. We use the 100-dimensional vectors from the GloVe 6B release (Pennington et al., 2014). The averaged document vectors are fed into Logistic Regression trained on NYT training examples.

Word2Vec trained on AG News. A 100-dimensional Word2Vec model is trained on AG News text (Mikolov et al., 2013). Its embeddings are then averaged over NYT document tokens, and Logistic Regression is trained on the resulting NYT document vectors.

Word2Vec trained on NYT. A separate 100-dimensional Word2Vec model is trained **only with NYT training-split text**. Document-vector construction and classifier training follow exactly the same pipeline as the AG News experiment. This strict setting prevents information leakage from validation or test partitions.

3.3 Fine-tuned BERT

We fine-tune `google-bert/bert-base-uncased` as a sequence classifier (Devlin et al., 2019). Input text is tokenized with a maximum sequence length of 64, with truncation and padding applied. The model is trained for exactly three epochs. We monitor performance on the validation set during training. The final model is evaluated once on the held-out NYT test set.

4 Implementation Details

All experiments adopt unified label encoding and the fixed 80/10/10 dataset split. Random seed 42 is fixed across all modules for reproducibility. Feature-based methods employ Logistic Regression from scikit-learn with default hyper-parameters. Static word embeddings are fixed to 100 dimensions. Word2Vec models are implemented via gensim using Skip-gram (sg=1). For BERT fine-tuning, learning rate is set to 2×10^{-5}. All BERT training runs on CPU hardware.

Package versions used in this project: scikit-learn==1.6.1, gensim==4.4.0, transformers==4.57.6, torch==2.8.0, nltk==3.9.2. Word2Vec hyper-parameters: vector_size=100, window=5, min_count=2, negative=5, sg=1, epochs=5, workers=4, seed=42.

5 Results

Table 1 summarizes the test-set scores from experiment outputs. Scores are rounded to four decimal places.

5.1 Results Analysis

Among three bag-of-words baselines, Word Frequency achieves the best test-set performance, with Accuracy of 0.9886 and Macro-F1 of 0.9725. It outperforms Binary BoW by +0.0043 in Accuracy and +0.0097 in Macro-F1, indicating term occurrence counts bring additional useful lexical signals for this news classification task. TF-IDF yields intermediate performance. This answers our first

Method	Accuracy	Macro-F1
Binary BoW + Logistic Regression	0.9843	0.9628
Word Frequency + Logistic Regression	**0.9886**	**0.9725**
TF–IDF + Logistic Regression	0.9860	0.9672
GloVe 6B (100d) mean + Logistic Regression	0.9773	0.9498
Word2Vec trained on AG News (100d) + Logistic Regression	0.9729	0.9399
Word2Vec trained on NYT (100d) + Logistic Regression	0.9747	0.9451
BERT-base-uncased fine-tuning	0.9703	0.9354

Table 1: Performance on the held-out NYT test set.

research question: frequency-aware lexical features outperform simple binary presence features.

For static embedding-based approaches, pre-trained GloVe obtains the highest scores. NYT-trained Word2Vec slightly outperforms Word2Vec trained on out-of-domain AG News corpus, showing limited benefit from in-domain corpus. However, all averaged embedding methods are inferior to sparse bag-of-words representations, which addresses our second research question.

Fine-tuned BERT-base-uncased obtains the lowest overall result (Accuracy 0.9703, Macro-F1 0.9354). A major reason is the 64-token truncation discards context information of long news articles. Even with contextual modeling capacity, BERT cannot surpass simple lexical baselines under this configuration, answering the third research question.

Across all seven models, the majority *sports* class achieves near-perfect F1, while minority *business* and *politics* categories suffer worse performance, which highlights the necessity of Macro-F1 given severe class imbalance in NYT dataset.

6 Discussion

Sparse bag-of-words representations preserve explicit lexical signals and are computationally cheap. Word frequency performs best because raw term counts capture topic-related word patterns for news classification. Nevertheless, bag-of-words ignores word order and semantic relationships between words.

Averaged static embeddings produce compact dense document representations and can share semantic knowledge across similar words. GloVe outperforms Word2Vec variants here, benefiting from large-scale general-domain pre-training. NYT-trained Word2Vec is marginally better than AG-News Word2Vec, confirming modest in-domain corpus benefits. However, simple averaging erases word-order information and weakens discriminative topic signals.

BERT supports contextual token representation. However, our BERT result is worse than feature baselines. The 64-token upper limit truncates long news articles and removes meaningful content. Besides, CPU-only execution restricts hyper-parameter exploration. Complex representation models do not guarantee superior performance for every dataset under fixed experimental constraints.

7 Limitations

This work has several notable limitations. First, we only adopt one random 80%/10%/10% data split with seed 42; we cannot measure performance variance across multiple random seeds. Second, static embedding representations rely on simple averaging, which discards word-order and local contextual information. Third, BERT input length is restricted to 64 tokens, leading to information loss for long news articles. Fourth, all BERT experiments run on CPU without systematic hyper-parameter sweep. Fifth, the NYT dataset suffers severe class imbalance dominated by the *sports* category. Even with Macro-F1 as evaluation metric, model predictions are still biased toward the majority class. Note that NYT Word2Vec is strictly trained on training-only text so there is no data leakage.

8 Conclusion

This paper compares three families of text representations for news classification on the imbalanced NYT dataset: sparse bag-of-words features, averaged static word embeddings, and fine-tuned BERT. Word frequency combined with Logistic Regression achieves the best overall performance, reaching 0.9886 Accuracy and 0.9725 Macro-F1 on test data. TF-IDF and binary BoW follow. Among embedding-based approaches, pre-trained GloVe performs better than Word2Vec trained on AG News or NYT corpus. Fine-tuned

BERT-base-uncased yields the lowest scores under our configuration.

Future work could evaluate multiple random data splits for stability assessment, enlarge BERT maximum input length to retain full news context, and conduct systematic hyper-parameter search.

Reproducibility Checklist

The NYT dataset is shuffled and split into 9157 / 1145 / 1145 (80%/10%/10%), and this fixed split is reused for every method.

All reported scores are measured only on the held-out NYT test set containing 1145 samples.

Dataset statistics, package versions, Word2Vec hyper-parameters and CPU hardware are documented in Section 4.

The BERT-base-uncased model is fine-tuned for exactly three epochs with max sequence length set to 64.

Word2Vec trained on NYT uses only training-split text to eliminate data leakage risk.

Source code and execution instructions are provided in the submitted GitHub repository.

References

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